密码国门与自主图强

密码技术的国家战略安全之路



NSA后门事件

- Code snippets reveal deeply embedded surveillance mechanisms, bypassing encryption protocols.
- The exploit leverages vulnerabilities in widely used firmware, creating persistent access points.
- Data exfiltration channels operate with high levels of obfuscation, making detection extremely difficult.
- Global security experts analyze the systemic impact, calling for immediate infrastructure audits.



华为技术封锁

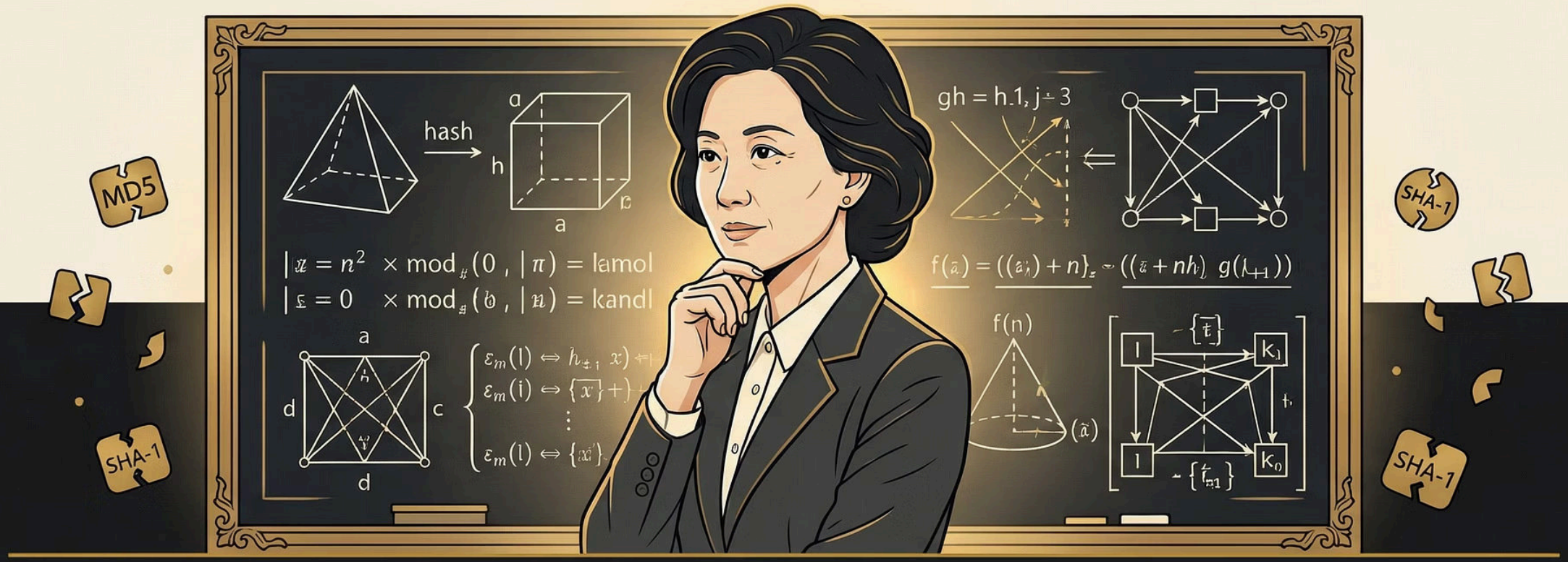


US Entity List: Restrictions prevent access to essential US technology and components. Impact on semiconductor and software supply chains.

Global Disruption: Broken links in the international market, affecting hardware and software development. The search for alternative solutions.

Technological Independence: Efforts to develop proprietary operating systems and chips. The rise of HarmonyOS and self-reliance initiatives.

王小云院士的科学壮举



1. Cryptographic Hash Functions: Understanding the Backbone of Digital Security.

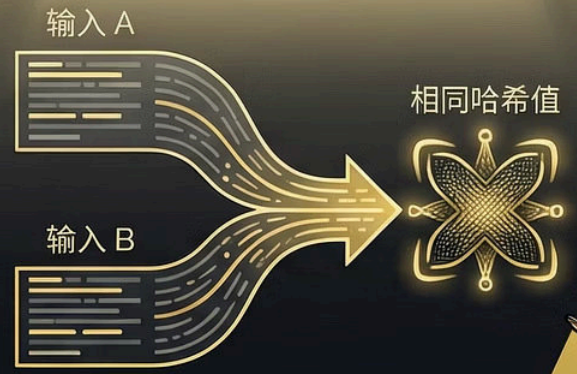
2. The Wang Attack: Breaking MD5 and SHA-1 Algorithms.

3. Impact on Global Cybersecurity: A Paradigm Shift.

4. Legacy and Future: Inspiring a New Generation.

破解MD5的历史时刻

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor inctidunt ut labore et dolore magna aliqua ut. enam.



Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor inctidunt ut labore et dolore magna aliqua ut. enam.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim

从破解到创造



- Foreign algorithm analysis and deconstruction.

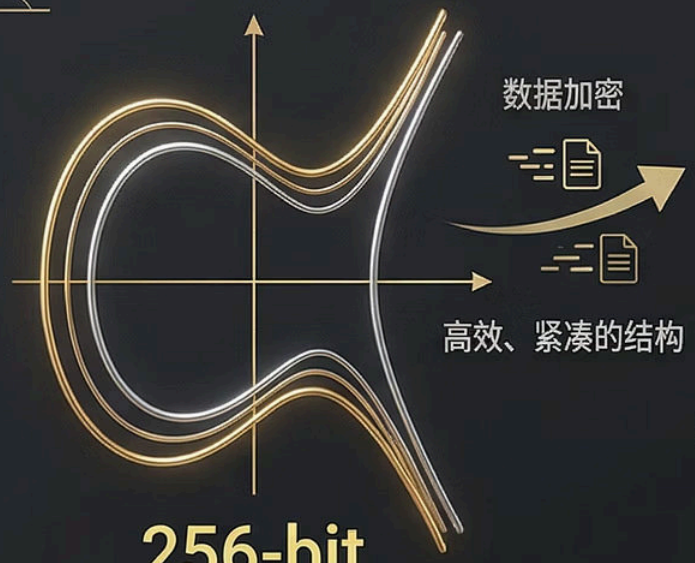


- Collaborative development and secure implementation.

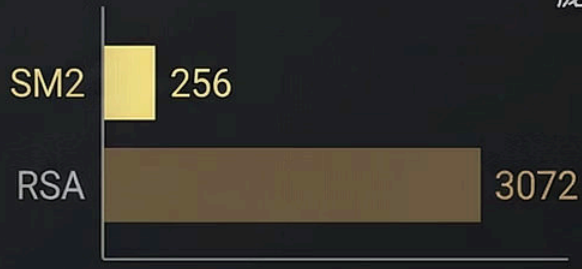


- Construction of new Chinese algorithm with SM ciphers.

SM2 vs RSA：椭圆曲线的优势









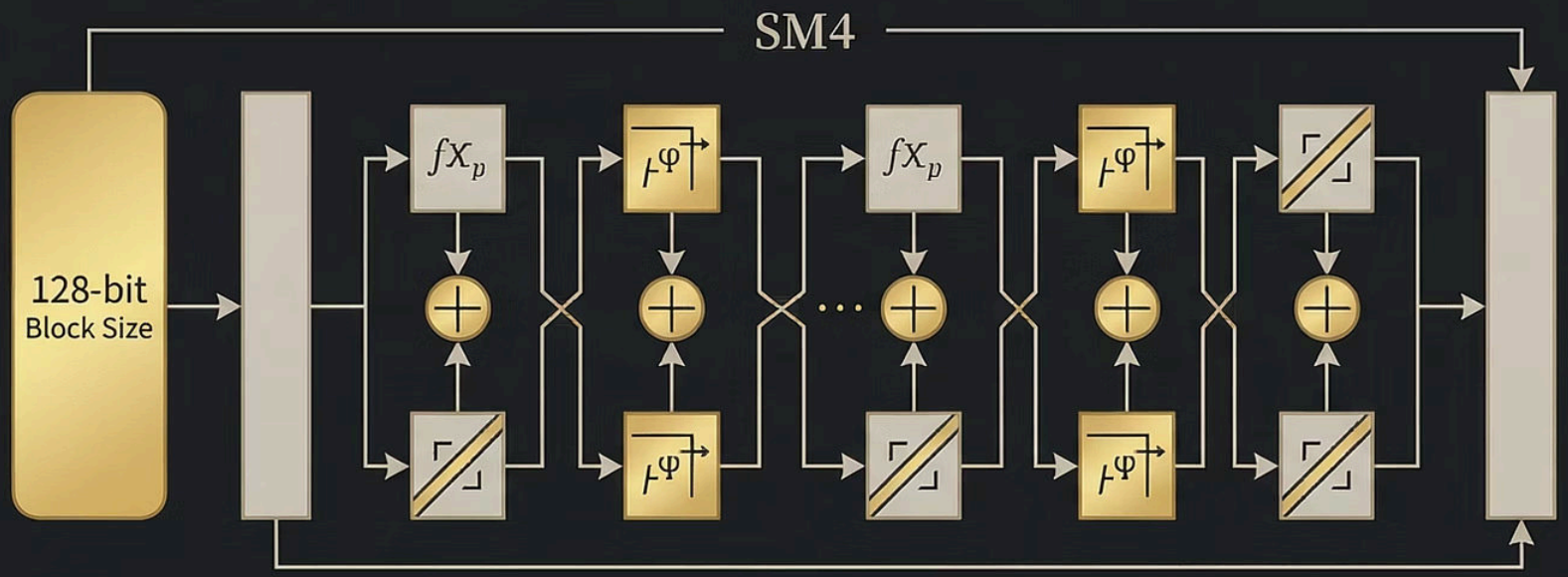


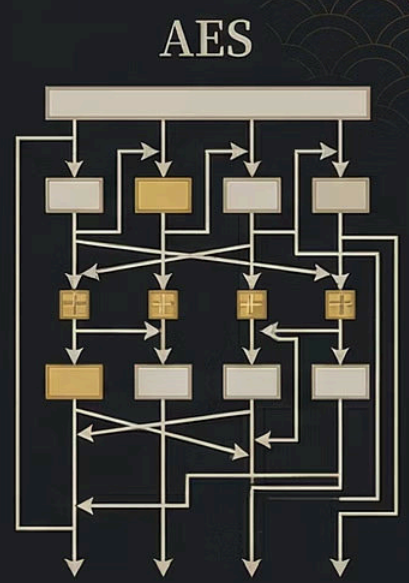
SM2 算法基于椭圆曲线加密，仅需 256 位密钥即可达到相当于 RSA 3072 位的安全性，在性能和效率上具有显著优势。

得益于更短的密钥长度，SM2 在签名和加**密速度**上可比同等安全强度的 RSA **快 10 倍以上**，显著减少计算资源消耗。

RSA 算法需要非常长的密钥来维持高安全性，导致计算开销大，尤其不适用于资源受限的物联网、移动设备等场景。

SM4 vs AES：结构的简洁之美





结构分析
Structure Analysis

SM4采用简洁的Feistel结构，轮函数设计精巧，实现高效非线性与线性变换。

Compared to AES, its structure is notably simpler and symmetrical.

简洁之美
Elegant Simplicity

对称的S盒和线性层设计，使得硬件实现更紧凑，代码更精简，展现出算法设计的优雅。

性能优势
Performance Metrics

在受限环境下表现优异，更快的时钟频率和更小的芯片面积。

High efficiency in resource-constrained environments.

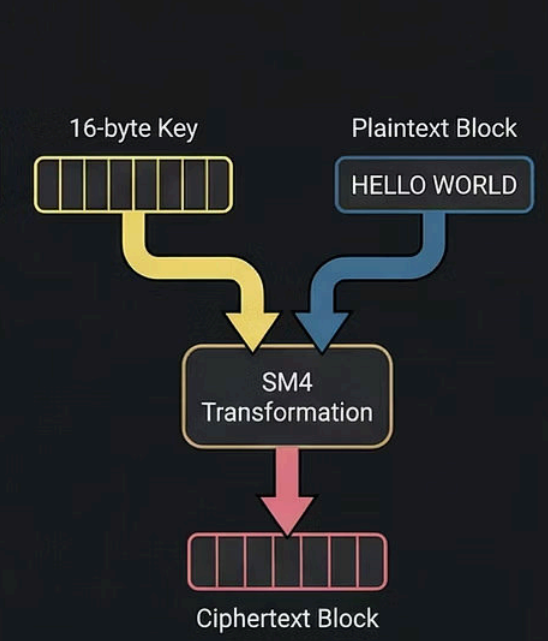
gmssl库实战演示

```python
from gmssl import sm4

# key initialization
key = b'0123456789abcdef'  # 16-byte key

cipher = sm4.SM4(key)
ciphertext = cipher.encrypt(b'HELLO WORLD')

# decryption decryption
decrypted_text = cipher.decrypt(ciphertext)
```



SM4 Key Initialization

The 16-byte required for b'0123456789abcdef' key initialization, encryption to systems noun requeras a 16-byte key.

Encryption Process

The function ~~tlhre~~ hora call to ciphertext = cipher. encrypt(!HELLO WORLD')

Decryption Verification

The reverse operation is func the necieus snomenis rellowing decryption for decrypted_text = cipher.decrypt(ciphertext).

金融IC卡中的国密应用



国密算法集成

- 安全芯片硬加密
- 密钥保护机制
- 数据完整性保障

跨境支付安全

- 国际标准兼容
- 支付全链路加密
- 风险实时监控

行业应用场景

- 公共交通
- 社保医疗
- 数字身份认证

5G通信的国密保障



国密标准应用

采用国密算法（SM2，SM3、SM4）确保数据机密性与完整性，构建安全可控的5G通信环境。

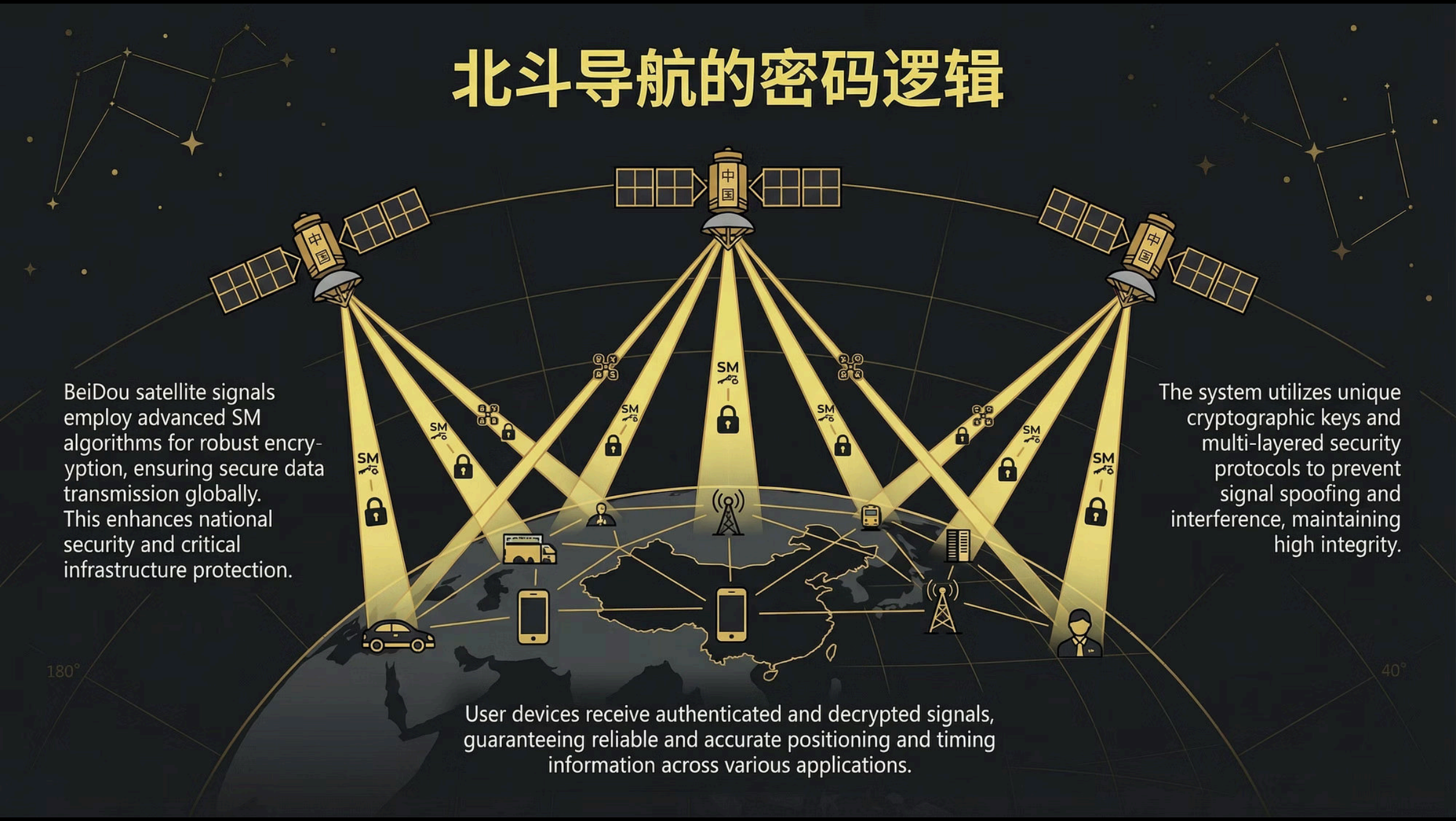
核心网安全架构

基于服务化架构（SBA），实现网络切片与边缘计算的安全隔离，保障关键信息基础设施。

身份认证与密钥管理

利用国密证书体系进行设备与用户的双向认证，加强密钥生命周期管理，防范未授权接入。

北斗导航的密码逻辑


BeiDou satellite signals employ advanced SM algorithms for robust encryption, ensuring secure data transmission globally. This enhances national security and critical infrastructure protection.
The system utilizes unique cryptographic keys and multi-layered security protocols to prevent signal spoofing and interference, maintaining high integrity.
SM
SM
SM
SM
SM
SM
SM
180°
40°
User devices receive authenticated and decrypted signals, guaranteeing reliable and accurate positioning and timing information across various applications.